

CO4 AT2 OUTPUTS

1.

```
er' '51035' '--' 'C:\Users\reddy\nlp\DSA0304-NATURAL-LANGUAGE-PROCESSING'
show
me
the
transactions
with
the
card
from
last
month

Possible Interpretations:
1. Transactions were made using the card.
2. Transactions are related to the card.

Subject-Verb Agreement: Correct

Improved Parser: Earley Parser
Reason: Handles ambiguity and long sentences efficiently.
PS C:\Users\reddy\nlp\DSA0304-NATURAL-LANGUAGE-PROCESSING> c::; cd 'c:\
```

2.

```
er' 51151' '--' 'C:\Users\reddy\nlp\DSA0304-NATURAL-LANGUAGE-PROCESSING\
Input: book a flight to Delhi with a window seat

Top-Down Parsing:
Starting from Sentence (S)
S -> Verb + Object + Details
Backtracking may be required for ambiguous parts.

Possible interpretations:
1. Book a flight to Delhi.
2. Book a flight with a window seat.
3. Book a flight to Delhi with a window seat.

Earley Parsing:
Handles multiple interpretations.
Handles incomplete input.
Uses dynamic programming.
Suitable for real-time voice assistants.

Comparison:
Top-Down Parser -> More backtracking
Earley Parser -> Better ambiguity handling
PS C:\Users\reddy\nlp\DSA0304-NATURAL-LANGUAGE-PROCESSING> c::; cd 'c:\Us
URAL-LANGUAGE-PROCESSING'; & 'C:\Users\reddy\AppData\Local\Python\python
c:\Users\reddy\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bu
er' '63346' '--' 'C:\Users\reddy\nlp\DSA0304-NATURAL-LANGUAGE-PROCESSING\
Ln 28, Col 67 Spaces: 4 UTF-8 CRLF { } Python 8
```

3.

```
er' 50514' '--' 'C:\Users\reddy\nlp\DSA0304-NATURAL-LANGUAGE-PROCESSING\CO4_AT2\Q3.PY'
Input Sentence:
The doctor who reviewed the patient last week
recommends starting medication and scheduling a follow-up visit in Chennai.

Step 1 - Tokens:
['The', 'doctor', 'who', 'reviewed', 'the', 'patient', 'last', 'week', 'recommends', 'starting',
medication', 'and', 'scheduling', 'a', 'follow-up', 'visit', 'in', 'Chennai']

Step 2 - CFG Structure:
S -> Subject + Verb + Object
Subject -> The doctor + Relative Clause
Relative Clause -> who reviewed the patient last week
Verb -> recommends
Object -> starting medication + scheduling follow-up visit

Step 3 - Feature Structure:
Subject: doctor
Number: Singular
Verb: recommends
Agreement: Correct
```

